



Data handling in data fusion: Methodologies and applications

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ABSTRACT

The use of data fusion methodologies has increased at the same rhythm as the capability of modern analytical laboratories of measuring sample from multiple sources. Almost all data fusion strategies can be grouped into three levels, they fuse the data differently with the sole aim of obtaining a better response (qualitative or quantitative) than that obtained by the instruments individually. One of the major key points for the data fusion methodologies to succeed is the understanding of the data structure obtained from a particular instrument. This point is not exhaustively commented in the literature focused on data fusion, sometimes paying too much attention to the algorithms instead. This manuscript explains data fusion from the structure of the different data obtained by different analytical platforms. Special attention will be given to the nature of the data and the relationships between the samples and the variables, as well as within the variables.

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1. Introduction

Modern laboratories are multi-analytical platforms where there is normally the possibility of making different analyses on the same sample. Different analytical instruments, especially those treating the sample in a non-destructive manner, can offer diverse chemical information from the same sample. By themselves, the instruments give very rich analytical information. Nevertheless, appropriately mixing the analytical signals could reveal certain synergic answers that could not be seen otherwise [1]. This ability promoted the idea that a sample might be better known or characterized by jointly analyzing its multi-platform analytical information. This idea is what we know as data fusion (DF) strategies.

Therefore, DF aims to obtain improved models, either exploratory or predictive (quantitative or classification), from the joint analysis of the diverse data obtained from multiple sources [2]. This aim implies or, better said, considers that the outcome of the analysis will be more reliable and that different phenomena occurring in the samples will be more understandable if the data are analyzed together. The interest is, most often, focused on assessing the commonalities and differences between the different data sets, also taking into account their linking relation.

Merging the data coming from different analytical platforms is not an obvious step due to the different nature and dimensionality of the data. Moreover, the merging affects the intrinsic nature of the final data and the data analysis model to be performed. Different analytical platforms require different merging strategies as well as different data analysis strategies.

Generally speaking, all types of data could be merged somehow. Nevertheless, the strategy to follow strongly depends on the relationship between the samples and variables, the relationship within the variables obtained with one instrument, and the relationship between the data coming from different instruments. That is, knowing the structure of the data is essential to set the most adequate merging scenario.

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Nomenclature

x (1×1)	plane lowercase. Scalar. Zeroth-degree tensor
$\mathbf{x}$ ($1 \times N$)	bold lowercase. Vector. First-degree tensor. Dimensions in italic uppercase
$\mathbf{X}$ ($M \times N$)	bold uppercase. Matrix. Second-degree tensor. Dimensions in italic uppercase
$\underline{\mathbf{X}}$ ($M \times N \times J$)	bold uppercase underscore. The dimensions of the tensors, M, N, J , in italic uppercase

DF cannot be understood without the active implication of machine learning/chemometrics strategies. Data are normally multivariate. Fusing multivariate data will generate multivariate fused data. Therefore, multivariate data analysis strategies are used. Nevertheless, the multivariate strategies need to be adapted to the new fused-data structure created.

In the literature, we can find many manuscripts and reviews defining different strategies for fusing data and analyzing it towards the pursued target [1,3,4]. Nevertheless, the literature normally focuses more on the algorithmic part than the structural part of the DF. Therefore, this manuscript aims to emphasize the data structure and how they can be merged into different data analysis scenarios.

The manuscript is divided into the following sections: a first part focused on the structure of the data acquired by analytical instruments. The second part discusses the typical definitions of terms found in the literature (their similarities and differences will be revised). The third part puts into perspective the major strategies in DF. The last part of the manuscript shows an example where DF is applied at different levels to develop a classification model. The example shows the fusion process in detail, emphasizing the major and minor benefits and drawbacks of DF and how this fusion affects the performance of classification models and their validation [5].

2. Data generated by analytical devices. Structure

Information about a sample can be obtained from different types of instruments, measurement techniques, experimental setups, and other types of sources. Samples are often heterogeneous mixtures, and according to that, each instrument is aimed at detecting specific properties (or combinations). Considering the diversity and structural differences of data, the underlying challenge is to understand how datasets can be fused 1) to extract the combined relevant information and 2) not distort results due to an incorrect fusing method. Thus, understanding the type, structure, nature, dimensionality of the data; and the correlation between and within the samples and variables is essential.

Hence, developing a method based on multiset data processing implies, among the development of the method itself, an in-depth study of the properties and the characteristics of the obtained data to select appropriate preprocessing strategies and the most suitable algorithms for the chemometric resolution.

Data can be classified in many different manners. The most useful manner for understanding the final structure is considering the instrument with which it is acquired. Zeroth-order instruments produce a single response per sample. For each measurement, a zeroth-order tensor (i.e. a scalar) is obtained. Examples of these data are temperature, pH, or the concentration of a compound. Zeroth-order measurements for M samples can be grouped to form a column vector $\mathbf{x}$ ($M \times 1$) for each independent variable, and, thus,

a matrix $\mathbf{X}$ ($M \times N$) can be arranged measuring N independent variables (Fig. 1a).

First-order instruments are the ones measuring the properties of a sample in different dependent variables. Therefore, a vector (first-order tensor) is obtained for each measurement from an only sample. This kind of data can be acquired from analytical techniques such as chromatographic or spectroscopic, for instance, vibrational spectroscopy (mid-infrared (MIR), near-infrared (NIR), and Raman), ultraviolet–visible spectroscopy (UV–vis), nuclear magnetic resonance (NMR), X-ray spectroscopy, among the more reported. Depending on the purpose of the analysis, the vector can be treated as such, or zeroth-order information can be extracted from it (e.g. the area under the curve between specific variables or the peak intensity of a particular variable) (Fig. 1b).

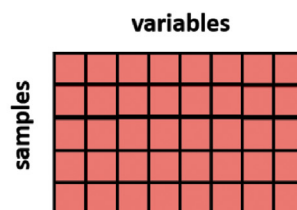
On the other hand, second-order or higher-order instruments give a full matrix (second-order tensor) for each sample, where the two directions of the matrix are dependent variables in de mode (Fig. 1c). On this basis, an alternative nomenclature has been developed in which data are named according to the number of ways (modes) of an array for a sample set. Thus, zeroth-order data is also one-way data; first-order is two-way, second-order is three-way, and so on [6]. Thus, it is often the case that having measurements of a completely different nature, a matrix $\mathbf{X}$ ($M \times N$) can be generated (Fig. 2a–f).

Fig. 2a shows that measuring the chromatographic profile at N elution times of M samples generates an $\mathbf{X}$ matrix. A matrix with the same number of dimensions could be obtained if, instead of elution time, the peaks are integrated and then, the areas of the peaks become the new variables (Fig. 2b). Even containing similar analytical information, both matrices should be treated and visualized as completely different data structures since the natural connection between variables in the first matrix (intensity at consecutive elution times) is lost in the second matrix. A similar situation can be seen in Fig. 2d (first-order instrument) and 2e (zeroth-order instruments). The four cases represent four different situations. Nevertheless, the common link between the latter four cases is the sample mode. For all of them, the samples remain the same, and their linkage point should be considered.

A completely different situation can be found in Fig. 2c and f where a matrix $\mathbf{X}$ can be obtained by measuring variables in M and N directions. Fig. 2c represents a monochannel image where an independent intensity value is measured in the spatial domain limited by the pixels in M and N . There, the relationship between the pixels is not from a chemical nature but a spatial one. Instead, measuring a continuous emission spectrum with N wavelengths at different M excitation wavelengths generates a matrix $\mathbf{X}$ where there is a natural linkage between the intensity measured at different wavelengths (second-order instruments) [7].

The same care must be taken with higher-order data (Fig. 2g–k). The complexity in the structure can be extrapolated from the previous cases where knowing the linkage between the variables is the key point to understand the nature of the data. For instance, measuring M samples in hyphenated chromatographic devices (e.g. gas chromatography with mass detection) generates a three-way structure where the variables are dependent on the N elution times and the K mass fragments (Fig. 2g). A similar structure is obtained with EEM fluorescence, where the M samples are measured at N excitation and K emission wavelengths (Fig. 2h).

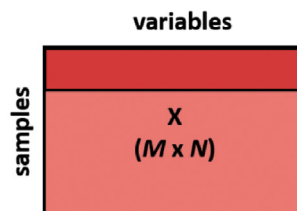
A different situation is found in Fig. 2j and k, where the nature of the variables differs in the final aim or the target of the problem to be solved. For instance, when batches are compared (Fig. 2j), there might be a need to establish a temporal dependency or study whether the batches belong to the same population. Similarly, in environmental exercises (Fig. 2k), the seasons might have

a) Zeroth-order dataOne sample
One variableMany samples
One variableMany samples
Many variables**b) First-order data**

One sample



Many samples

**c) Second-order data**

One sample

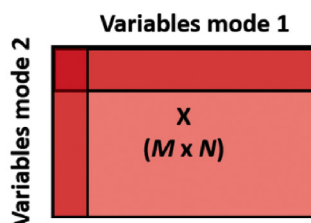


Fig. 1. Structure of data generated by different devices. a) Zeroth-order data, b) First-order data, and c) Second-order data.

an intrinsic temporal relationship. In any case, the treatment of those data cubes (Fig. 2g, h, j, k) could be done by natural three-way methods or by unfolding the cubes into matrices. This point will be addressed further when we talk about low-level strategies.

A particular case is represented by hyperspectral imaging data (Fig. 2i). Even though three-dimensional data is obtained for each sample, there is no prior connectivity between the pixel directions since they should be considered as somehow independent sampling points. This assessment is not completely correct since there is a spatial relationship between them. Nevertheless, to what mathematical decomposition concerns, it is normally assumed that the spatial relationship is temporarily lost due to an unfolding procedure, converting the three-dimensional data into a matrix X with dimensions $(MN \times K)$. Several constraints can be added to the mathematical calculations to preserve the spatial relationship. Nevertheless, this is out of the scope of this review, and the reader is encouraged to check the provided references for further information [8].

3. Data fusion. Multiset. Multi-block. Matrix augmentation. Matrix couple. What is data fusion, and what is not data fusion?

Considering the structure of the data to be fused, the first important definition is the mode. A mode is defined as one of the categories according to which the data can be ordered [9]. The more frequent situation reported in the analytical chemistry field has been different sets of variables measured for the same set of samples having the observation mode the common link. For instance,

data could be generated by two different spectroscopic techniques, such as ultraviolet–visible (UV–vis) and near-infrared (NIR) spectroscopies. In this case, spectral profiles are multivariate (as the responses are acquired at several wavenumbers) and multimodal (the two different spectroscopic techniques represent the modes) (Fig. 3a).

Similarly, data matrices can have the common mode in the measured variables having different samples. For instance, when multiple batches of an industrial process produce data under different processing conditions (Fig. 3b) [4,10].

The corresponding final arrangements of the fused datasets are not simply multivariate but also can be named by different terms, such as multiset/multi-block since it consists of multiple matrices linked or coupled. Thus, the integration of all datasets/data-blocks acquired on the same system or phenomenon from different sort of instrumental techniques, and multimodal, understanding the data coming from multiple sources as modalities (e.g. each technique as a modality). Moreover, the term concatenation is particularly applied to data blocks arranged in an augmented multiset sharing the mode of the sample. Then, the merged matrix is modelled as a single block, and the contribution of the individual blocks is not contemplated [2].

On the other hand, it is also possible that the data matrices would have both modes in common. For example, a series of LC-DAD spectra have as common modes retention time and the spectral mode. Independently of the applied technique, a data set can be arranged into a) a single three-dimensional array or cube, stacking each matrix on top of each other $\underline{X}$ ($M \times N \times J$); b) unfolding or vectorizing the original matrices, which indicates the

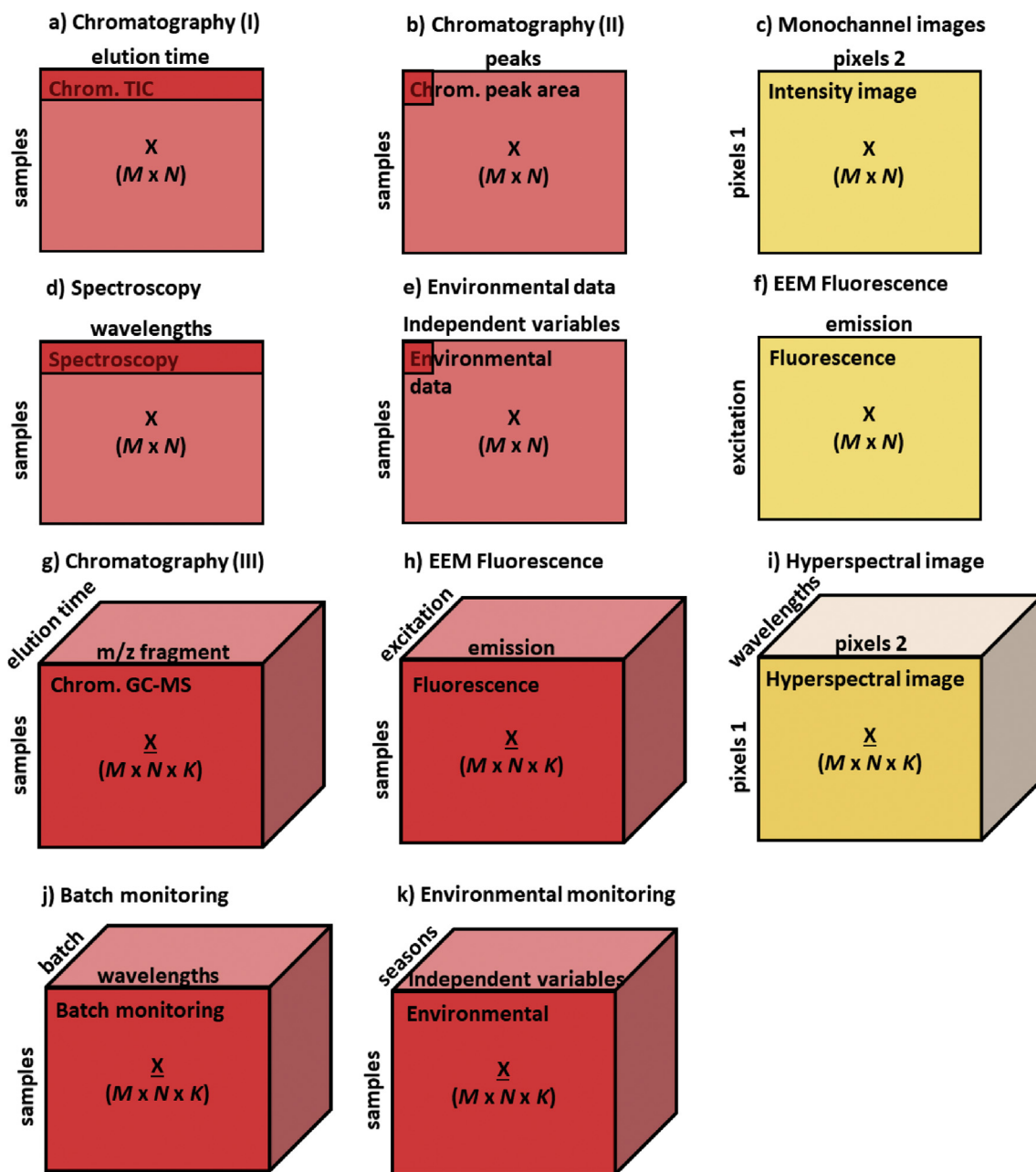


Fig. 2. Examples of how a matrix X ($M \times N$) can be obtained by different instruments. a) represents chromatographic profiles, while b) represents the integrated areas of peaks in chromatographic profiles; c) represents a monochannel image; d) a common spectroscopy case; e) arrangement of independent variables in environmental data, and f) the matrix obtained in excitation-emission fluorescence. Also, several three-way structures generated by different instruments: g) hyphenated chromatographic device; h) Excitation-emission fluorescence; i) Hyperspectral image; j) Batch monitoring with spectral devices, and k) environmental monitoring.

process of reshaping a matrix in a vector, and is performed by concatenating two-way arrays (matrices); or c) an augmented matrix can be generated by arranging the instrumental response matrices in the following ways: row-wise direction (one matrix below the other) (Fig. 4a), or column-wise direction (one matrix next to the other) (Fig. 4b). Notwithstanding, from this kind of data, a single data block is acquired. Highlighting DF as an approach that integrates multiple data blocks/platforms, the analysis on a single data set is not conceptualized as a DF. Otherwise, any of these three types of data arrangements can be adopted for multi-blocks analysis. In fact, b and c options are mainly implemented as a strategy to connect datasets with tensors in a different order since the different typologies of these structures define the optimal way to do the fusion.

4. Data fusion levels and strategies

In analytical chemistry, the possible ways of fusing the data are generally grouped into three general strategies, depending on the combination in which DF occurs: low-level (LLDF), if the raw data is used as an input in the DF procedure; mid-level (MLDF) when the extracted features of the data are used; and high-level (HLDF) when the data are combined at the classification/prediction decision level. However, a wide range of different strategies can be adopted within each level depending mainly on the dimensionality and structure of data, and even multiple levels can be combined. Fig. 5 displays the schematic representation of data analysis workflow building multi-platform models by applying LLDF, MLDF and HLDF strategies and showing the main steps of each.

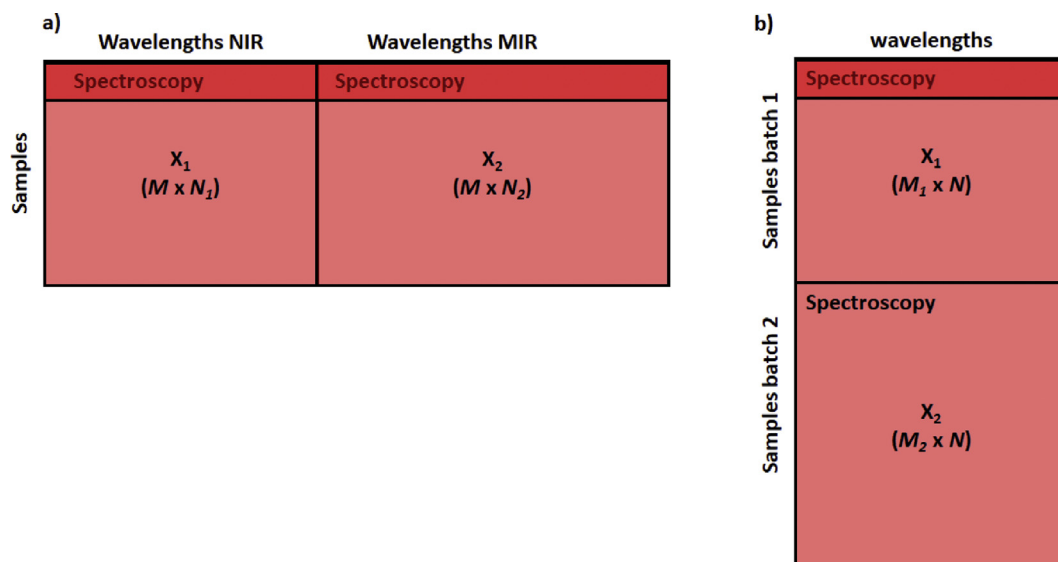


Fig. 3. a) Fusion of the spectroscopic data generated by two different spectroscopic techniques for the same set of samples; b) Fusion of the spectroscopic data generated by the same spectroscopic technique for different samples.

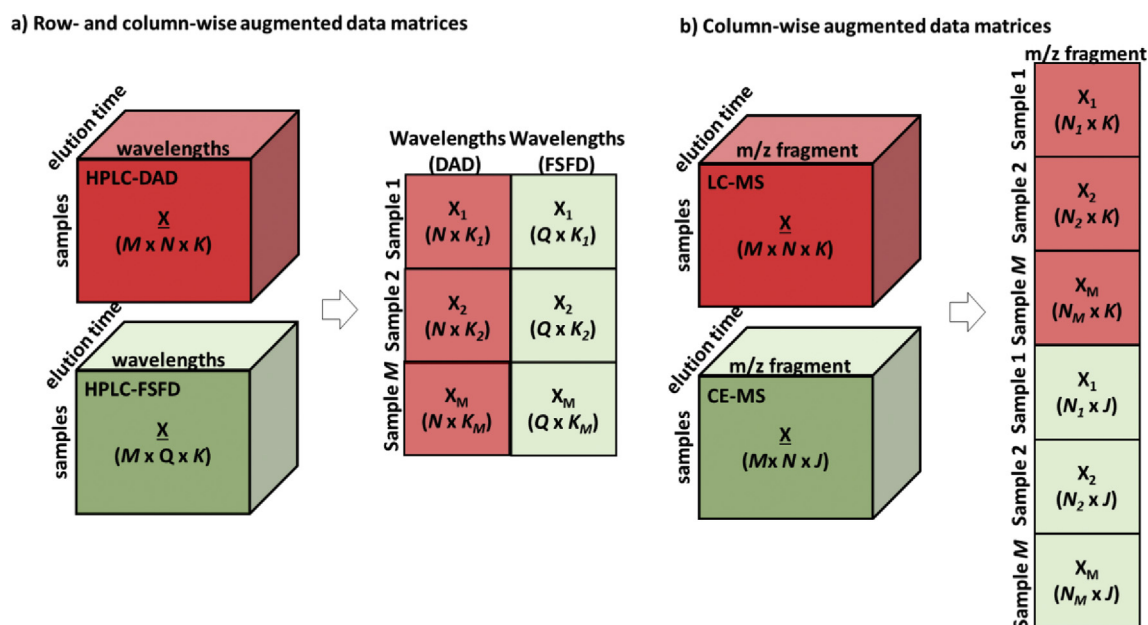


Fig. 4. a) Fusion of second-order data generated by the same hyphenated chromatographic techniques with different detectors for the same set of samples; b) Fusion of second-order data generated by the two different hyphenated chromatographic with the same detector for the same set of samples.

4.1. Low-level data fusion

LLDF is the most straightforward and conceptually simple way of fusing data. It is normally used as a first approach when there is more than one dataset available. It comprises the coupling of datasets in the samples or the direction of the variable (Fig. 3a), depending on the common mode, by directly placing the datasets next to the other (e.g. study of different variables of the same set of samples) or coupling them across the variables (Fig. 3b), (e.g. the same variables are measured on samples belonging to different batches) [11]. LLDF can also be called data augmentation or multi-block analysis, especially in multivariate curve resolution (MCR) scenario or even unfolding in hyperspectral literature.

LLDF keeps all the original information from different sources. However, it normally contains noise and redundant information, which affect the precision of modelling and calculation. Hence, prior and post steps of data preprocessing are usually required to solve this problem (Fig. 5a).

Concerning the “prior” preprocessing or pretreatment in LLDF, it is really important to have each data in the mode “perfect-to-be-used”, which means that each dataset should be previously pre-processed in the same way as when it is used individually before the combination of the fusion matrix. Thus, the highest amount of noise and uninformative variables should be removed before the DF, and undesirable problems such as baseline and scattering should be corrected. In other cases, mainly with zeroth-order, different classical variable normalization methods (e.g. autoscaling or Pareto

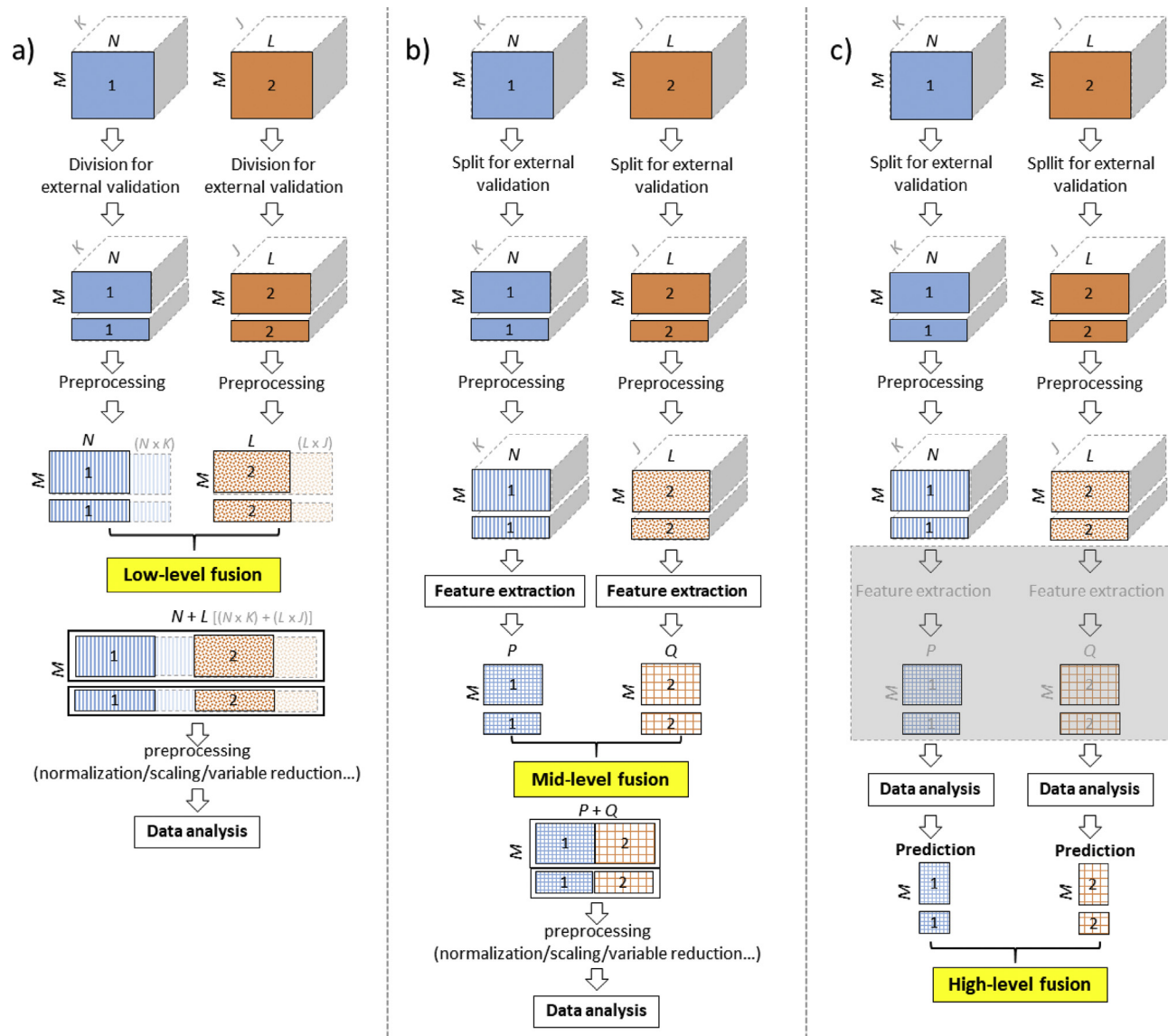


Fig. 5. Schematic representation of data analysis workflow representing the a) low-, b) mid-, and c) high-level data fusion.

scaling) are applied as pretreatment due to the different nature of the variables considered.

In general, the individual preprocessing methods will be the one more according to the nature of the data. An example of LLDF of data from a very different nature is seen in the paper published by Gonçalves et al. (2020) [12], in which the physicochemical results were autoscaled, the UV–vis spectra were smoothed, and the baseline was corrected, and the first derivative was applied to the NIR spectra for the baseline correction. After individual preprocessing, the UV–vis, NIR, and GC data blocks were also individually normalized. This individual normalization is a step that is not always performed first to the DF. On the contrary, it is preferable when the measured features show a difference in variances or are concerned with very different measured entities or units [12]. Variable selection or reduction is advised in LLDF to avoid the predominance of one large matrix over the others, which may decrease the performance of the model and avoid useless and time-consuming calculations [1]. It should be highlighted at this point that if the variable selection is made “manually”, i.e. by looking at the data or due to previous knowledge of the variables (e.g.

elimination of the spectral band related to water due to it is not relevant for the study), it is still LLDF. It could be said that you are not extracting information from variables but removing uninformative or not relevant variables. Even more, it could be possible to divide the set into different subsets (for example, regions of the spectra) and concatenate them in multiple combinations [13].

In contrast, when a method of data reduction or variable selection is applied, it could be considered LLDF or a MLDF, depending on the level applied [14,15]. ANOVA is also another approach of variable selection or reduction of the concatenated data that could be considered in order to improve the performance of the LLDF model [16]. Other approaches could also be considered in LLDF, not only focused on facilitating the computation but also on retaining the relevant information from each signal type (Table 1).

Concerning the preprocessing of the fused data, it is of utmost importance when the raw data have very different magnitudes. It is required that the data from different sources have comparable scales, including the unbalanced number of variables, which should be compensated in order to prevent one block from being dominant in the subsequent data analysis [1,5]. For this purpose, block-

Table 1

Tabular overview of recent application fields and analytical data fusion approaches of the different technologies and chemometric methods used in researches.

DF level ^a	Data type	Technique ^b	Fusion strategy ^c	Objective	Matrix	Results	Ref.
LLDF	0th order	E-nose + E-tongue	Concatenation preprocessed RD/selected variables	Classification	Food	Results improve by DF	[16]
	0th + 1st order	UPLC + NIR + Physicochemical parameters	Concatenation preprocessed RD	Exploratory analysis	Food	Results improve by DF	[26]
	0th + 1st order	Physicochemical parameters + GC + UV –vis + NIR	Concatenation preprocessed RD	Exploratory analysis	Food	Good results-no comparison	[12]
	0th + 1st order	LIBS + NIR	Concatenation preprocessed RD + selection of variables	Regression	Food	Results improve by DF	[28]
	0th + 2nd order	E-nose + reflectance Hyp. + Transmittance Hyp. NIR, MIR, RAMAN	Concatenation preprocessed RD/selected variables	Classification	Food	Good results-no comparison	[30]
	1st order		Concatenation selected variables	Classification	Food	Not improve	[13]
	1st order	UV–vis, Synchronous fluorescence + NIR	Concatenation preprocessed RD	Classification	Food	Results improve by DF	[15]
	1st order	MIR + NIR	Concatenation preprocessed RD	Classification	Food	Not improve	[32]
	1st order	LIBS + WDXRF	Concatenation preprocessed (Zeroth-order) RD	Regression	Food	Results improve by DF	[27]
	1st + 2nd order	GC-IMS + FT-MIR	Concatenation unfolded data	Classification	Food	Results improve by DF	[17]
	1st + 2nd order	NIR + F	Concatenation preprocessed RD/unfolded data	Regression	Food	Results improve by DF	[18]
	2nd order	LC-DAD + LC-FSFD	Concatenation of preprocessed RD reduced to 0 th order	Regression	Materials	Results improve by DF	[21]
	2nd order	CE-MS + LC-MS	Concatenation preprocessed RD	Identification	Metabolomic	Results improve by DF	[22]
	0th + 1st + 2nd order	Conductivity and pH + NIR + EFM	Feature extraction and concatenation with 0 th order data	Monitoring	Environmental	Good results-no comparison	[41]
MLDF	1st order	E-senses data (e-nose, e-tongue and e-eye)	Concatenation of PCA scores	Characterization	Food	Results improve by DF	[40]
	1st order	FT-IR + Raman	Concatenation of PCA scores	Classification	Clinical	Results improve by DF	[37]
	1st order	FT-IR, UV, and ICP- AES combined	Concatenation of PCA scores	Classification	Food	Results improve by DF	[36]
	1st order	NIRS + ATR-FT-IR + ICP-OES	Concatenation of PCA scores	Classification	Food	Results improve by DF	[35]
	1st order	Visible + NIR + NMR	Concatenation of MCR and PCA scores	Exploratory analysis	Food	Results improve by DF	[34]
	1st order	NIR + UHPLC-HRMS	Concatenation of PCA scores	Exploratory analysis	Metabolomic	Results improve by DF	[33]
	1st + 0th order	E-nose + FT-NIR	Concatenation of PLS scores	Exploratory analysis	Food	Results improve by DF	[49]
	1st + 0th order	Colorimetric + E-nose + NIR	Concatenation of VIP	Regression and Classification	Medicinal materials	Results improve by DF	[50]
	1st + 0th order	FT-NIR + CVS	Concatenation of scores	Classification	Food	Results improve by DF	[39]
	2nd order	Raman + Fluorescence	Concatenation of PCA scores	Classification	Materials	Results improve by DF	[38]
Multilevel	1st + 2nd order	MIR, NIR, EFM, NMR	LLDF: concatenation of preprocessed RD; MLDF: concatenation of feature extraction (PCA, PARAFAC MCR) and concatenation.	Classification	Food	Results improve by DF	[5]
Comparative LLDF + MLDF	1st order	HS-MS, FT-MIR, UV–vis	LLDF: concatenation of preprocessed RD; MID: concatenation of PLS scores	Calibration	Food	Results improve by DF: LLDF = MLDF	[61]
	1st order	NMR two different samples	LLDF: simply concatenation; MID: concatenation of VIP	Classification	Clinical	MLDF better than LLDF	[43]
	1st order	FT-MIR + ICP-AES	LLDF: concatenation of preprocessed RD; MLDF: concatenation of PCA scores	Classification	Food	MLDF better than LLDF	[14]
	1st order	¹ H NMR + LC-MS + GC-MS		Classification	Food	MLDF better than LLDF	[44]

(continued on next page)

Table 1 (continued)

DF level ^a	Data type	Technique ^b	Fusion strategy ^c	Objective	Matrix	Results	Ref.	
Comparative LLDF + HLDF	1st order	NIR + ATR-FTIR	LLDF: concatenation of preprocessed RD; MLDF: concatenation of VIP variables	Classification	Food	MLDF better than LLDF	[45]	
	1st order	UV–vis + NIR	LLDF: concatenation of preprocessed RD; MLDF: concatenation of PCA scores	Regression	Chemicals	LLDF better than MLDF	[48]	
	1st order	FT-IR + Raman	LLDF: concatenation of preprocessed RD; MLDF: concatenation of PCA scores	Regression	Clinical	MLDF better than LLDF	[46]	
	1st order	UV–vis + UHPLC/Q-TOF-MS	LLDF: concatenation of preprocessed RD; MLDF: concatenation of PLS scores	Regression	Food	MLDF better than LLDF	[47]	
	1st order	ATR-FTMIR + NIR	LLDF: concatenation of preprocessed RD; MLDF: concatenation of VIP variables or PCA scores	Regression	Food	Results improve by DF: MLDF better than LLDF	[31]	
	1st order	Mid-wave IR + Long-wave IR	LLDF: concatenation of preprocessed RD; MLDF: concatenation of PCA scores	Regression	Materials	LLDF better than MLDF	[60]	
	1st + 0th order	ATR-FTIR + physicochemical variables	LLDF: concatenation of preprocessed RD; MLDF: concatenation of VIP variables or PLS-DA scores	Classification	Food	Results improve by DF: LLDF better than MLDF	[42]	
	0th order	Stable isotope, multielement, fatty acids	LLDF: Concatenation; HLDF: ratio modulation	Classification	Food	Not improve by LLDF, Improvement with MLDF	[29]	
	1st order	(HT)GC-FID + HPLC-CAD	LLDF: concatenation of preprocessed RD; HLDF: Fuzzy operators-majority voting	Classification	Food	Results improve by DF: LLDF better than HLDF	[55]	
	2nd order	CE-MS + LC-MS	LLDF: concatenation by column-wise augmentation; HLDF: data integration	Identification	Metabolomic	Results improve by DF	[22]	
	Comparative MLDF + HLDF	1st order	¹ H NMR + UV–vis	MLDF: concatenation of iPLS selected variables; HLDF: Fuzzy Aggregation Connective Operators	Classification	Food	Results improve by DF	[52]
		1st order	FT-Raman + NIR	MLDF: concatenation of selected variables; HLDF: Fuzzy Set- Majority Voting	Classification	Food	Results improve by DF: MLDF = HLDF	[51]
	Comparative LLDF + MLDF + HLDF	1st order	MIR, NIR, RAMAN, PTR-MS, E-nose	LLDF: concatenation of preprocessed RD; MLDF: Concatenation of PCA scores; HLDF: Bayesian consensus and Majority voting	Classification	Food	Results improve by DF: HLDF the best	[57]
		1st order	NIR + MIR	LLDF: concatenation of preprocessed RD; MLDF: hierarchical (concatenation of PLS-DA scores) or multi-block (MB-PLS-DA); HLDF: Fuzzy and Majority voting	Classification	Food	LLDF not improve; HLDF the best	[54]
		1st order	FT-IR + UV	LLDF: concatenation of preprocessed RD; MLDF: Concatenation of PCA scores; HLDF: Fuzzy	Classification	Food	LLDF and MLDF not improve; Improve by HLDF	[53]

2nd order	EEM + CE-DAD	LLDF: concatenation unfolded RD; MLDF: concatenation of PCA scores or extracted features; HLDF: Majority voting and Bayesian consensus	Classification	Food	Results improve by DF: HLDF the best	[19]
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Notes: ^a LLDF: low-level data fusion; MLDF: mid-level data fusion; HLDF: high-level data fusion. ^b UPLC:: NIR:: Hyp.: hyperspectral; GC: gas-chromatography; FT-MIR: Fourier transform-mid infrared spectroscopy; UV-vis: ultraviolet-visible spectroscopy; LIBS: laser-induced breakdown spectroscopy; WDXRF: wavelength dispersive X-ray fluorescence; GC-IMS: gas-chromatography-ion mobility spectroscopy; F: fluorescence spectroscopy; LC-DAD: liquid chromatography-diode array detection; LC-FSFD: liquid chromatography-fast scanning fluorescence detection; CE-MS: capillary electrophoresis-mass spectrometry; ICP-AES: inductively coupled plasma-atomic emission spectrometry; NMR: nuclear magnetic resonance; CVS: computer vision system; HS-MS: headspace-mass spectrometer; UHPLC/Q-TOF-MS: ultra-high performance liquid chromatography/quadrupole time-of-flight mass spectrometry; (HT)GC-FID: high temperature gas chromatography-flame ionization detector; HPLC-CAD: high-performance liquid chromatography-charged aerosol detector; PTR-MS: proton transfer reaction-time of flight-mass spectrometry; CE-DAD: capillary electrophoresis-diode array detection. ^c RD: raw data; PCA: principal component analysis; VIP: variable importance in projection; PLS-DA: partial least squares-discriminant analysis; IPLS: interval partial least-squares regression.

normalization of the data is usually applied in the fused matrix. Different preprocessing or normalization can be applied according to the nature of the raw data matrices, being one of the most used block-scaling (Table 1). Thus, block-scaling aims to equalize variance equal to one, but the variance ratio between any two variables inside a single block is preserved. Other block-scaling methods have been evaluated in other works, such as mean-centering, weighting, auto-scaling, the inverse of the sum of squares, root square scaling, log scaling and second derivatives between others (Table 1).

LLDF is mainly applied in zeroth- and first-order data due to the facility in concatenation. For second-order data, other levels are mostly preferred. Instead, an unfolding step is usually performed before the concatenation. This fact implies losing the meaning of the variables when the models are performed, making the interpretation of results more difficult. Some examples of the concatenation of the unfolded second-order data with other data of the same or different nature could be found in the literature [17–19].

Another possibility to concatenate data blocks of different nature can be into a single structure augmented in the row and/or column mode. In this context, MCR-ALS has allowed the joint analysis of multiple datasets from different samples (experiments) or techniques or different samples and techniques simultaneously in an LLDF [20]. The current strategies have been mainly implemented in multiset data provided by hyphenation from separation techniques being typical examples of identification and quantitative analysis of compounds. Thus, DF can be done in many different modalities. For instance, a chromatographic run with diode array detection (DAD) and fast scanning fluorescence detection (FSFD) in tandem provides a row-wise augmented matrix, with a single elution mode then, a row- and column-wise augmented multiset is obtained when chromatographic runs from different samples are analyzed together (Fig. 4a) [21]. Otherwise, multisets acquired from chromatographic and electrophoretic runs with MS detection can be merged in a column-wise augmented data matrix (Fig. 4b) [22]. However, before merging, a series of steps is mandatory to overcome some problems such as shifts and distortions in chromatographic profiles, different time channels among detectors, and scaling factors to balance intensity differences among intensity signals.

To preserve multi-way structures in an LLDF when arrays present different numbers of modes, for instance, first- and second-order data, other kinds of strategies have been accomplished. In this context, sequential and orthogonalized PLS regression to multi-way arrays (SO-N-PLS) has been implemented for predictive and classification analysis [23]. Moreover, the coupled tensor and matrix factorization approach has been reported for unsupervised analysis in the DF field [24]. Despite both approaches have been scarcely implemented, these reports have demonstrated that, also in DF context, better results can be achieved when natural dimensionality of the multi-block data is maintained.

It is often assumed that the result of the fusion of two or more different data matrices will promote more informative results than the analysis of the individual matrices [15,25–29]. However, this improvement with LLDF is not always observable. It could occur that one single technique provides almost perfect results, and the combination with another that provides worse results would lead to the deterioration of results [17]. Moreover, it has also been observed that the fusion of two data matrices from different sources increases the accuracy of the models, but the fusion of a third one does not increase it, or even make it worse. This means that not always the fusion with more matrices will lead to an improvement of the models from the individual ones or the fusion of only certain blocks [30–32] due to the addition of factors like a large amount of interference, disturbance, redundant or noisy and

useless information. Therefore, the benefit of LLDF strongly depends on the information represented by each data, that is the complementarity or synergy of them, as well as the application of good preprocessing methods of the data prior and post the fusion.

4.2. Mid-level data fusion

MLDF is based on a two-step methodology where it first extracts relevant features from each data source separately and, secondly, these outputs are concatenated to build a single matrix to be processed (Fig. 5b).

The first step normally involves calculating latent variables or selecting features independently extracted from each analytical data matrix (Fig. 5b). A straightforward model interpretation in terms of original variables is accomplished from variable selection methodologies. However, a more demanding assessment of the results in terms of the raw variables is required to ascertain the relationship between the salience of each feature in the final model and its corresponding pattern in terms of original variables.

The reduction of the dimensions can be carried out in different ways. Principal component analysis (PCA) has been the most popular method applied to compress first-order data or unfolded second-order data. Then, the extracted PCA scores from different matrices are concatenated to model fused data [33–40], avoiding to a greater extent the application of variable selection algorithms [2].

MLDF is the DF strategy that poses more challenges for second-order data, where different strategies for data compression, extraction or reduction have been explored for improving outcomes [2]. According to the data structure, when second-order data can be arranged in a data cube, it can be decomposed by parallel factor analysis (PARAFAC) into trilinear components where scores (first mode loadings) can be used to build a fused dataset [41]. If the presence of artefacts hampers for a direct trilinear decomposition, methods like unfolded MCR-ALS or PARAFAC2 can be used to extract the scores of the components [34,41].

Obviously, this approach also solves the issue of concatenating data blocks of a different order. In this sense, several attempts have been addressed to evaluate the combination of first- and second-order data provided by multiple platforms since data vary in structure, size or scale [5,39,41,42].

The performance of MLDF applications has commonly been compared to LLDF and single models achieving, in the majority of the cases, improved results in the final model that could be associated with the features reduction step since the noninformative variance is removed [14,19,43–46].

Regarding some disadvantages, MLDF performs more exhausting methodologies because it requires applying as many selection algorithms or building as many models as data sets. Moreover, selecting the proper number of components (principal components, latent variables, etc.) is another crucial step in this DF level since useful information can be left out in the residuals if a proper data reduction is not performed [42,47,48].

The data block scaling is less critical in MLDF with respect to LLDF, but choosing the dimensionality of the models (extracting the relevant number of latent variables) block is introduced. Supervised methods can also carry out the dimension reduction. For instance, PLS (or another multivariate supervised approach) can be applied on each matrix and then, scores corresponding to these components are concatenated. Subsequently, PLS, or another supervised technique can be performed to the concatenated score matrix to obtain the final model [49,50]. When a supervised method is applied, a particular situation can be considered since their application at this stage may be confused with a HLDF approach. However, only the features, not the model results, are fused. Thus, the

DF level could not be considered strictly mid owing to extracted features containing information about the final model sought [2].

4.3. High-level data fusion

HLDF approaches are less used in the analytical chemistry field, probably because they require a major complexity than LLDF and MLDF approaches. This level encompasses straightforward development since only the outputs coming from different models are fused (Fig. 5c).

According to the literature, there are no references of works that apply directly and only HLDF on their data without trying a lower level first (Table 1). HLDF is a suitable strategy to integrate heterogeneous information, in such a way that each data block is treated independently and, then, the outcomes (predictions) are fused to analyze them as a single block (Fig. 5c) [2]. In HLDF a new matrix containing as many rows as samples and as many columns as needed to fulfil the outcomes from the different analysis of the independent data blocks is created. This data matrix is then analyzed to get a fused final prediction. The outcomes might be quantitative or qualitative (class information) indistinctly and, therefore, special care must be put in the construction of the matrix. For example, the dummy-matrix rules must be followed when classification outcomes are fused and, then, proper normalization must be performed to equalize the importance (weight) of all new variables created.

HLDF has mostly focused on classification issues. Several examples can be found in the literature fusing outcomes from different analytical platforms like NIR and FT-Raman [51], NMR and UV–vis [52], FTIR and UV [53], NIR and MIR [54], HPLC-CAD and (HT)GC-FID [55], to highlight the significant improvement of the classification result of the HLDF respect to those obtained from individual models (Table 1).

Several strategies have been proposed for analyzing the classification results that involve the majority voting and Bayesian consensus with discrete probability distributions [2,56], among others [19,57]. When dealing with the fusion of quantitative values, different operators can be calculated and adapted to provide the final combined outcome, such as the average, the maximum, or the minimum value, and the weighted average based on predictive indicators or the product fuzzy aggregation connective operators. Fuzzy aggregation connectives is a method that can be used to model a HLDF system of the output classification from domain data [58] easily. A majority vote rule has been used to obtain common criteria for processing the information provided by the operators. Several applications of these methodologies have been reported in the literature [19,51,54].

4.4. Comparison between the three strategies

By comparing LLDF with the other superior two levels, MLDF and HLDF, there is no general rule of improvement of the results as you move to higher levels. However, this improvement with higher levels strategies usually occurs in most of the literature found (Table 1) [31,43,45,46,59]. It could be highlighted that there have been some cases in which LLDF has been able to provide better results than MLDF approaches [42,48,60], and sometimes the two levels provided similar results [61,62], but most of the studies showed better results by MLDF than by LLDF [14,44–46].

If LLDF provides better results, it could be explained by the fact that the original information from both data blocks is better maintained. Moreover, the drawbacks discussed above about LLDF are compensated because it does not have the difficult problem of needing an optimal extraction of features that MLDF has. However,

as with other DF levels, its usefulness is not a general rule, and it should be evaluated in each case.

HLDF does not need to adjust an adequate scaling due to each model being fitted independently with its best scaling. However, a disadvantage of HLDF is that the order of combining the predictions obtained affects the final decisions. Sometimes, there could be confusions in the assignment of the level of DF used when these algorithms are applied in different data analysis levels. Hence, in the case of Zhao et al. [46], and Nunes et al. [42], the concatenation of VIP variables with other raw data was considered LLDF, whereas some other authors considered it as MLDF instead [31,44,47]. Moreover, the DF level is not defined sometimes because it could be difficult to be assigned [52,63].

The application of validation also has some differences between the DF levels. Nevertheless, they all have something in common: the error propagation will be smaller if the external validation is done from the beginning. In the case of cross-validation, it should be made when each model is performed. Hence, it is very simple in the LLDF because there is only one classification or prediction model after the concatenation. In contrast, in the MLDF it should be made in each model that is performed to select the best options for each one, leading to a better final MLDF model (i.e. reducing the intermediate errors will reduce the final error). Finally, in HLDF, as each matrix is individually modelled, the cross-validation is made in each model, being intrinsically performed in the final model.

Regarding the external validation, it should be again performed at the beginning in all the levels. This means that the dataset should be first split into two matrices, a train set and a test set, and then, each one should be preprocessed, concatenated, modelled, and so on separately. Thus, in LLDF, each matrix should be first split into two, the train and the test sets, being the same samples retained in each one for each dataset, and then each one should be preprocessed, concatenated and normalized equally but independently. On the other levels, it should also be done for each matrix initially, and the same train and test sets should be maintained for the different models. This fact can be seen in the workflow of a recently published work [19].

5. Example

An example of the complexity/versatility of the DF approaches has been recently published [5]. In this work, the vinegar samples were measured by different analytical platforms to classify sixty-five kinds of wine vinegar according to their protected designation of origin (PDO). The analytical platforms were MIR, NIR, multidimensional fluorescence spectroscopy (MFS) and proton nuclear magnetic resonance (^1H NMR). Classification models obtained from individual techniques were compared with those obtained by DF in order to find the optimized PDO differentiation.

As reported earlier, the four platforms gave four different data structures. Three of them (MIR, NIR and ^1H NMR) in the shape of a matrix $\mathbf{X}$ (samples $\times$ wavelengths) and a last one (MFS) in the shape of a three-dimensional data structure $\mathbf{X}$ (samples $\times$ excitation wavelengths $\times$ emission wavelengths). The specific details about the acquisition of each dataset can be found in the literature [5]. In all these spectroscopic techniques, samples were directly analyzed without pretreatment in random sequence at room temperature ($21 \pm 2^\circ\text{C}$) and by duplicate.

The data analysis workflow followed is shown in Fig. 6, including: 1) building of separate classification models using data coming from each analytical technique (except for NIR and MIR that were previously fused and then considered as one single matrix) (Option 1-individual classification); 2) all the instrumental signals were processed by means of different DF strategies (Option 2-MLDF classification).

The validation procedure consisted of splitting the samples into two groups employing the Duplex algorithm: a training set of forty-seven samples and a test set of eighteen samples.

5.1. Preprocessing

Concerning MIR data, the raw spectra were mean-centered after removing some uninformative regions (a 65×310 data matrix). Concerning NIR data, some regions related to low signal/noise ratio and the strong combination band of O-H from water were removed and, the obtained matrix of 65×1299 was preprocessed by smoothing (Savitzky-Golay filter, 7 points window and second-order polynomial degree) to reduce random noise, followed by standard normal variate (SNV) to correct additive scattering. Besides, the spectra were mean-centered before any analysis. Then, both preprocessed data sets were concatenated and normalized by block scaling and mean-centering in order to equalize variance and compensate for the different measuring scales and variability of each technique. This concatenation was considered a LLDF fusion step.

Regarding EEM data, EEMs were preprocessed in order to avoid noisy and noninformative areas by selecting shorter spectral ranges, obtaining a second-order data matrix of $65 \times 87 \times 246$. Then, this cube was corrected for Rayleigh and Raman scattering, removing and replacing the scattering areas with PARAFAC while it was multi-way centered on the first mode prior to the individual classification analysis. Finally, concerning ^1H NMR dataset, some spectral ranges were removed because they were uninformative or contained residual water signals, obtaining a first-order data matrix of $65 \times 21,240$. Then, all spectra were aligned using *icoshift*, whereas weighted least squares (WLS) were used for baseline correction.

Finally, different decomposition methods such as PCA, MCR or PARAFAC were also applied according to the type of dataset to reduce the data and obtain the relevant features, which were then used for individual classification or DF models (Fig. 6).

5.2. Classification methods

Methods like linear discriminant analysis (LDA) on the X-scores calculated by Partial least squares-discriminant analysis (PLS-DA) or N-way Partial least squares-discriminant analysis in the case of a three-way array (NPLS-DA) were applied for the individual classification models as well as for the MLDF classification. The minimum classification error rate in cross-validation (Venetian blinds, seven splits) was used to assess the number of latent variables, i.e. components of the PLS-DA/NPLS-DA models.

5.3. Data analysis and results

As occurs with preprocessing, the only way to know which decomposition method or which fusion strategy is the optimal one is to look into previous knowledge provided in the literature or test different combinations. The results obtained by the individual classification models (option 1) and the mid-level classification strategy (option 2) were assessed and compared. These classification results obtained by the individual (option 1) and mid-level PLS-DA models (option 2) are shown in Table 2.

5.3.1. Option 1. individual classification models

Regarding the first option (option1-individual classification), in the case of MIR and NIR individual data, the preprocessed matrices of the individual blocks were first concatenated to obtain a new matrix (LLDF), which was block-normalized and mean-centered, and used for developing a named "individual" PLS-DA classification model.

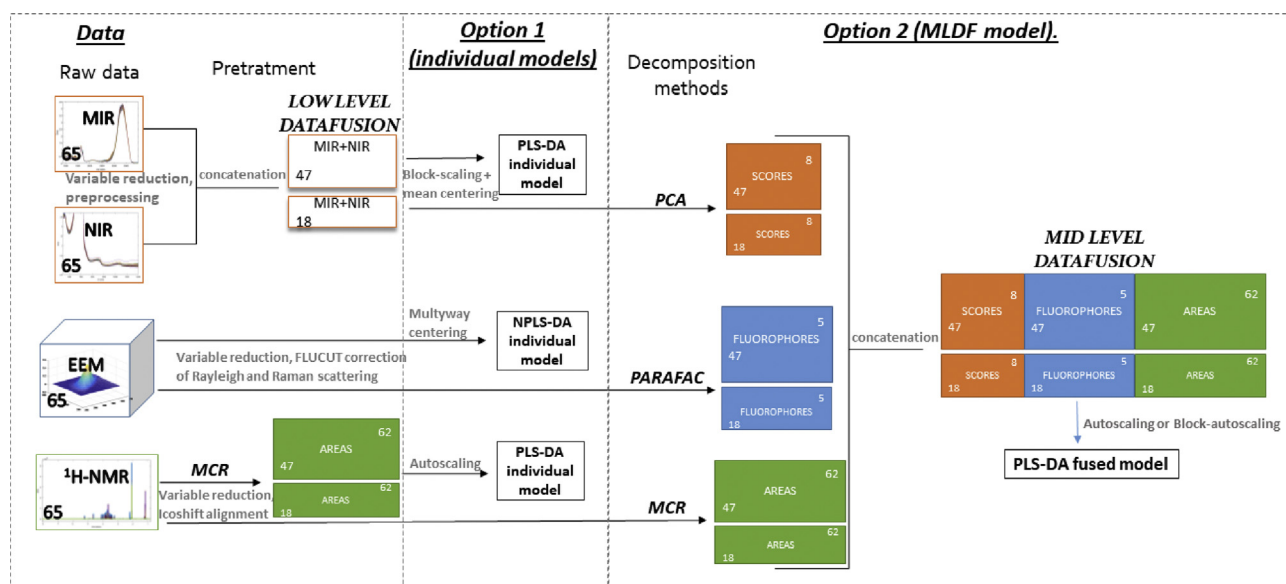


Fig. 6. Workflow analysis scheme modified from a previous work [5].

Table 2

PLS-DA results obtained for each individual data block and for low-level and mid-level fused dataset with different scaling procedures.

	Data	Classification method	Pretreatment	LVs ^a	% Corrected classified					
					Train			Test ^b		
					C	J	M	C	J	M
Option 1 (individual models)	MIR + NIR	PLS-DA	Block Scaling + Mean Centering	10	90.0	85.0	79.2	100	87.5	62.5
	EEM	NPLS-DA	Mean centering	12	66.7	95.0	75.0	50.0	100	83.3
	NMR	PLS-DA	MCR + autoscale	7	100	97.5	91.7	100	75.0	75.0
Option 2 (MLDF model)	Fused	PLS-DA	AUTOSCALE	6	100	100	91.7	100	100	100
	Fused	PLS-DA	AUTOSCALE + GS	7	100	97.5	91.7	100	100	100

Note: C = "Condado de Huelva" PDO; J = "Vinagre de Jerez" PDO; M = "Vinagre de Montilla-Moriles" PDO.

^a LVs number determined on the basis of minimum RMSECV with Venetian blind cross validation^b Independent test set, average correct classification rate for 5 random training/ test splitting is reported.

This model did not show satisfactory classification results regarding the percentage of error obtained in the train and test sets (Table 2). A classification model by each technique was performed individually, obtaining even worse classification results than the ones obtained by LLDF (data not shown).

Regarding the preprocessed EEM data array, the selected approach, according to the classification results, was to build an individual classification model using NPLS-DA (option 1 in Fig. 6). The classification results obtained (Table 2) were only better than those obtained by NIR-MIR for one of the PDOs (J), but they were worse for the rest. Moreover, PARAFAC was applied in order to extract the relevant fluorophores from the EEM. The PARAFAC factors obtained were also used to build an individual PLS-DA classification model, but the results obtained were even worse than those obtained by directly developing a NPLS-DA model with the EEM matrix (data not shown). This agrees with other authors [64] which showed that models built using second-order data and maintaining the three-way structure, especially in fluorescence data result in being more specific and sensitive than those built with first-order data.

Finally, for ¹H NMR dataset, after proper preprocessing, MCR was applied to resolve the chemical components, which peak areas were then used for building a PLS-DA classification model (option 1 in Fig. 6). The detailed MCR process was explained in the

mentioned paper of this example [5]. As we previously mentioned with the other spectroscopic matrices, although we selected MCR as the procedure to extract the relevant information because it was the one that provides the best classification results (Table 2), the raw spectra could have also been used for classification modelling. However, it needs to be properly scaled before analysis to avoid selecting the most abundant metabolites as significant. Therefore, it needs many preprocessing steps (normalization, Pareto scaling, group scaling and mean-centering), being a more complex procedure that did not provide better results. Moreover, the high spectral complexity and the richness of information it provides make the use of a chemometric method more adequate for the release and extract the meaningful information hidden in the NMR spectra.

5.3.2. Option 2. mid-level data fusion strategy

On the other hand, in the case of mid-level strategy (option 2 of Fig. 6), the concatenation of three extracted matrices was performed. The first matrix was obtained by applying PCA with 9 principal components (PCs) on the concatenated MIR and NIR preprocessed data, and the score vectors were used as MIR/NIR features to build the fused dataset. Moreover, it should be also mentioned that the concatenated NIR-MIR matrix was also directly used for developing the MLDF PLS-DA model, but the classification

results obtained (data not shown) were worse than those obtained after data reduction by PCA. This could be explained by reducing the spectra into the most relevant information of the classes (which is included in the principal components), delete noise, and uninformative data that worsen the classification. Moreover, for option 2, another possible alternative approach was considered by applying PCA to the separate MIR and NIR spectral data and then using the extracted features then in the MLDF (fused with other data matrices). In this case, this alternative yielded very similar results (data not shown). The second matrix concatenated in the MLDF (option 2) was built with the relevant information extracted by applying PARAFAC to the preprocessed EEM data array (scores of fluorophores). For that, a PARAFAC model based on five factors, with the constraint of non-negativity in both modes (concentration and spectral profiles), was built, selecting the proper number of factors [5]. The latter matrix, concatenated in the MLDF (option 2) consisted of the resolved peak areas, obtained from MCR-ALS application on the preprocessed ^1H NMR dataset.

Therefore, in this example, the final fused array was assembled using the 8 PCA scores from MIR and NIR concatenated matrix, the 5 factors from the PARAFAC model of EEM data, and the peak areas of the 62 resolved components by MCR-ALS of ^1H NMR data (Fig. 6). This fused matrix needed to be also scaled prior to classification analysis since the extracted features had different numerical characteristics. Although there are many different preprocessing tools, in this study only two were assessed, being the two most used in the literature: autoscaling and block-autoscaling.

As it can be clearly observed in the percentage of correct classification (Table 2), the MLDF approach (option 2), with any of both normalizations, improved the discrimination of PDO wine vinegars, providing more efficient differentiation rates in the training stage than the models based on single methods (with % of errors of even 34% in some PDOs in the training set, and up to than 50% of errors in the test set). Based on these outcomes, it can be concluded that, although ^1H NMR models were promising for a PDO classification objective, the single analytical methods by themselves seemed to be inappropriate to solve the proposed complex classification problem. In contrast, it could be clearly seen that MLDF models lead to excellent classification rates in the test set (100%), conversely to individual data from which a bad classification has been achieved (Table 2). Thus, the two scaling procedures furnished very similar PLS-DA classification rates, and only the number of latent variables, in addition to the classification rate in cross-validation, were different. Moreover, the performance of MLDF allowed achieving synergies for an optimized differentiation of the PDO of wine vinegars. This could indicate that although a single instrumental fingerprint was not completely able to predict the high-complex samples under study correctly; the possibility of combining the information from the different instruments by means of MLDF was accurate enough for solving this issue.

6. Final remarks

The ever-increasing availability of analytical techniques in laboratories and industrial processes remains demanding advances in DF in an assortment of fields. In this review, several DF were presented, and their applications were demonstrated on different types of data. Considering the three DF levels (LLDF, MLDF and HLDF), LLDF and MLDF approaches have been the most implemented in the literature. In general terms, LLDF has been the first attempt from reviewed applications when data from different sources were available. On the contrary, when data were very different in dimension, structure, size or scale, the MLDF has been the more used approach. Lastly, HLDF has been the least used. In general, it has also been observed that in most applications

considered in this review, better results in terms of classification and/or prediction improvement were yielded with DF with respect to the use of individual techniques.

From the different nature of data displayed in this review, the flexibility of DF has been revealed. In this sense, various applications demonstrated that, although diverse data types can be encountered in analytical chemistry, the same kind of DF strategy can be applied to analyze them. Otherwise, different type of DF strategies can be applied for the same multiset/multi-block. In this context, the next obvious question arises: Which strategy is the best for DF. As shown in this manuscript, the answer depends on the motivation and goal of the study and the acquired data, i.e., a priori best strategy cannot be envisaged. There is a lack of exploration and comparison of various DF approaches, mainly within the same level in most of the presented research.

New DF approaches are continuously emerging, like multi-level applications combining LLDA and MLDF, fusion of second and higher-order data, i.e. hyperspectral images or data from hyphenated techniques, and the combination of first-order data from evolving systems. Therefore, a more thorough study about optimally combining multiple data sources are necessary.

To conclude this manuscript, it is important to reflect on a key question before directing any research to a DF approach: DF is really needed? It again depends. If the aims of the study can be satisfactorily accomplished from a model built with a single instrument, a DF approach might not be needed. The availability of instruments should not drive the main aim of any research. The possibilities of having instrumental platforms to acquire data should not lead to apply DF methodologies directly. On the contrary, as researchers, we can hypothesize that an optimal outcome can be achieved by one instrument, or, if it is really needed, another available instrument could help improve the outcome of the experiment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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